

Keshav Dutt Kaushik

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SUMMARY

Backend-focused software engineer with 3+ years of experience building Python and Java services, with hands-on work in Retrieval-Augmented Generation (RAG) systems, vector databases, and document-processing pipelines. Comfortable across the full backend stack — API design, ingestion, embeddings, storage — and experienced shipping inside large enterprise codebases with significant legacy and migration work.

SKILLS

Languages: Python, Java, JavaScript

Backend: FastAPI, REST API design

AI / ML: RAG architectures, sentence-transformers, LangChain, embedding pipelines, semantic search

Data: Qdrant (vector DB), document parsing (PDF, DOCX, PPTX, XLSX, Markdown)

Tooling: GitHub Copilot, Swagger / OpenAPI, Git

EXPERIENCE

Cognizant

Software Engineer · August 2022 – Present · Pune, India

Generative AI Assurance Platform (Jan 2026 – Present)

- Migrating a production Flask backend to FastAPI while extending the API surface for new product features.
- Remediating data quality issues in the underlying database to support the modernized service layer.
- Coordinating with frontend developers to align the new React UI with revised backend contracts.

AILA — React Application Modernization (Jul 2025 – Nov 2025)

- Led component-level migration of a legacy React codebase to modern React, working through a prioritized component matrix.
- Remediated security vulnerabilities flagged during IP-readiness review and shipped the modernized application to delivery.

Knowledge Management Chatbot — RAG Platform (Dec 2023 – Oct 2024)

- Built ingestion and parsing pipeline for PowerPoint documents, including custom handling for tabular content embedded in slides.
- Designed embedding generation flow using sentence-transformers, with chunking via LangChain's recursive text splitter.
- Integrated **Qdrant** as the vector store and implemented CRUD APIs over the document and embedding lifecycle, exposed via **FastAPI**.
- Supported a unified ingestion model across PDF, DOCX, PPTX, XLSX, and Markdown formats, with later extensions for audio and video sources.

Document vs. Webpage Content Comparator (Oct 2022 – Nov 2023)

- Engineered the PDF parsing and text-structuring module of a Java **Spring Boot** comparator service using **Apache PDFBox**.
- Designed a heuristic paragraph-segmentation algorithm based on statistical line-length analysis to recover semantic blocks from raw extracted text.
- Contributed to comparison logic across word-, sentence-, and sequence-level mismatch detection in a cross-functional team.

EDUCATION

B. Tech, Computer Science and Engineering [2018-2022]